DANIEL KARIMI

SOFTWARE DEVELOPMENT ENGINEER

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ABOUT ME

Engineering student at **Universidad Politécnica de Madrid** (expected graduation 2025/2026), passionate about building scalable systems and integrating advanced AI capabilities into impactful business solutions. I possess a strong foundation in **Object-Oriented Design (Java, Python, C++)**, Data Structures, and modern **Software Development Life Cycle (SDLC)** practices including CI/CD and unit testing.

My recent experience centers around engineering robust data pipelines and scalable backend inference services at CSIC, where I designed intelligent tools leveraging Python, APIs, and Deep Learning (Generative AI & Computer Vision). I am highly adaptable to modern technology stacks, with exposure to **React, Cloud computing concepts (AWS), and distributed applications**.

I am eager to apply my robust algorithmic background and software engineering toolkit as a **Software Development Engineer** at Amazon Business, helping scale GenAI promotional systems and revolutionizing B2B commerce platforms.

TECHNICAL STACK

SOFTWARE ENGINEERING

Java • Python • C / C++ • Object-Oriented Design • Data Structures • Algorithms • Full SDLC

CLOUD & WEB TECH

AWS (Familiarity) • React (Frontend) • RESTful APIs • Distributed Systems • Git & GitHub • CI/CD Workflows • Docker

MACHINE LEARNING

Generative AI Concepts • PyTorch / TensorFlow • Scalable ML Pipelines • Deep Learning • Optimization • Data Modeling

TOOLS & BEST PRACTICES

Code Reviews • Unit Testing & QA • Agile / Scrum • Operational Excellence • Design Patterns • Scripting (Bash)

SYSTEMS & OS

Linux / Unix OS • System Architecture • Networking (TCP/IP) • Hardware Prototyping • Automation Scripts

ENGINEERING FOUNDATIONS

Problem Solving • Quantitative Analysis • Signal Processing • Microprocessors • Scalability Design

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council)

PROFESSIONAL EXPERIENCE

Software & AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI) — Sep 2025 – Feb 2026

- Spearheaded the development of **scalable Python-based software pipelines** to process, normalize, and analyze large-scale scientific datasets, significantly reducing raw data processing overhead.
- Designed and implemented object-oriented architectures for Machine Learning inference, bridging the gap between scientific models (PyTorch) and high-performance execution.
- Participated actively in the **full software development life cycle (SDLC)**, driving requirements gathering, algorithmic design, robust implementation, and extensive testing for accuracy and functional validation.
- Optimized complex data processing algorithms to ensure faster execution speed and maintainability across distributed research environments.
- Documented codebase architectures, engaged in code planning, and employed Git for source control management, fostering operational excellence and reproducible results within a cross-disciplinary team.

Python | Object-Oriented Design | Software Architecture | PyTorch | Testing | Optimization

EDUCATION

Telecommunications Systems Engineering | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Programming & Software Engineering:** Deep understanding of Data Structures, Object-Oriented Programming (OOP), algorithmic complexity, and design patterns using **Java, Python, C, and C++**.
- **Networking & Systems:** Comprehensive study of TCP/IP architecture, scalable network protocols, distributed operating systems, and Unix/Linux environments.
- **Engineering Foundations:** Mathematical modeling, digital signal processing (DSP), and complex problem-solving methodologies applicable to large-scale backend systems.

Java | C/C++ | Python | Algorithms | Distributed Systems

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong foundation in mathematics and physics.

CERTIFICATIONS

AI and Productivity (Google): Efficient use of AI tools, automation workflows, and operational efficiency. •

Generative AI (MIT): Implementation of GenAI solutions, foundation models, and intelligent integration strategies. •

Internet of Things (MIT): Distributed architectures, sensory data collection, and integration of IoT solutions. •

Digital Transformation (MIT): Technology strategy, agile methodologies, and scalable business digitalization. •

Aptis ESOL B2 (British Council): Official English certification. Capability to interact in highly technical, English-speaking professional environments.

KEY PROJECTS

CellTracker (Software Framework) BACKEND SYSTEM

PYTHON • OBJECT-ORIENTED DESIGN • ML INTEGRATIONS

Algorithmic tracking system pending publication in scientific journal.

Architected and implemented an end-to-end processing pipeline integrating custom Machine Learning features with robust algorithmic tracking (Kalman Filters). Focused on code reusability, modularity, and object-oriented design principles to handle heavy computational loads.

- Engineered reproducible systems and automated testing workflows for scientific validation.
- Demonstrated strong capabilities in algorithmic optimization and data structure utilization to optimize execution times.

MigraCell Accelerator AUTOMATION APP

PYTHON • DATA PIPELINES • OPTIMIZATION

Automated analysis tool leveraging deep learning for data extraction.

Developed a scalable software solution that drastically reduces processing time (from hours to minutes) by automating data extraction and integration. Designed the application logic to be extensible and maintainable, adhering to professional coding standards.

- Wrote clean, efficient Python code aimed at long-term maintainability.
- Integrated deep learning modules (YOLOv8-seg) within traditional software pipelines, showcasing adaptability to modern AI tech stacks.